

Artificial Intelligence Lab 1

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Section: BSCS-4B

Task 1

Create an array of all even integers from 200 to 240 using NumPy.

In [1]: `import numpy as np`

```
even_numbers = np.arange(200, 241, 2)
```

```
print(even_numbers)
```

```
[200 202 204 206 208 210 212 214 216 218 220 222 224 226 228 230 232 234
 236 238 240]
```

Task 2

Generate an array of 10 random numbers from a standard normal distribution.

In [2]: `import numpy as np`

```
random_numbers = np.random.randn(10)
```

```
print(random_numbers)
```

```
[-1.14758707 -0.06590169 -0.40888425  0.40732666  0.57987522 -0.44932772
 -0.43350844  1.49280351  0.49043376  0.88515695]
```

Task 3

Plot two lines and set line markers.

In [3]: `import numpy as np`

```
import matplotlib.pyplot as plt
```

```
x = np.arange(0, 11)
```

```
y1 = 2*x + 1
```

```
y2 = np.cos(x)
```

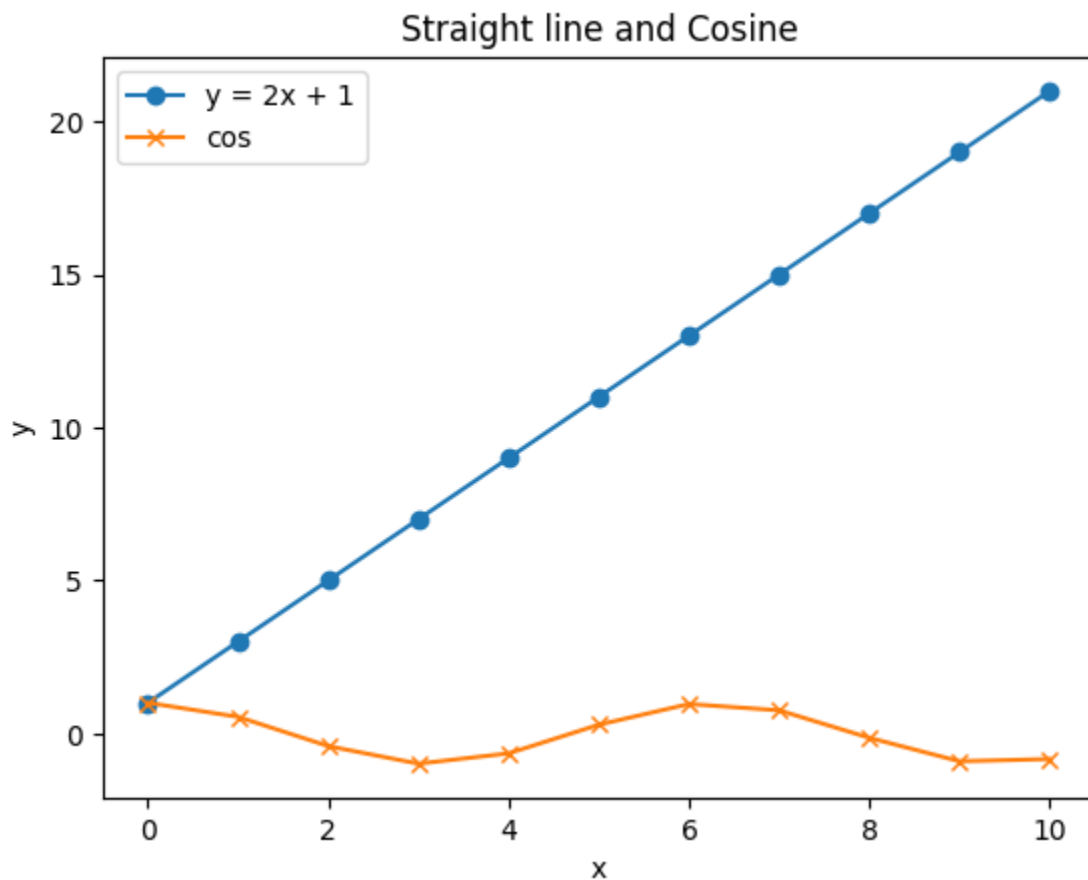
```
plt.plot(x, y1, marker='o', label='y = 2x + 1')
```

```
plt.plot(x, y2, marker='x', label='cos')
```

```
plt.xlabel("x")
plt.ylabel("y")
plt.title("Straight line and Cosine")

plt.legend()

plt.show()
```



Task 4

Change the datatype of a given Series.

```
In [4]: import pandas as pd

data = pd.Series([100,200,300,400])

print("Original Data Series:")
print(data)

data = data.astype('float64')

print("After changing datatype:")
print(data)
```

```
Original Data Series:
0    100
1    200
2    300
3    400
dtype: int64
After changing datatype:
0    100.0
1    200.0
2    300.0
3    400.0
dtype: float64
```

Task 5

Delete DataFrame rows based on a given column value.

```
In [5]: import pandas as pd

data = {
    'col1': [1, 2, 3, 4, 4],
    'col2': [4, 5, 6, 7, 8],
    'col3': [7, 8, 9, 0, 1]
}

df = pd.DataFrame(data)

print("Original DataFrame")
print(df)

new_df = df[df.col1 != 3]

print("New DataFrame")
print(new_df)
```

```
Original DataFrame
   col1  col2  col3
0     1     4     7
1     2     5     8
2     3     6     9
3     4     7     0
4     4     8     1

New DataFrame
   col1  col2  col3
0     1     4     7
1     2     5     8
3     4     7     0
4     4     8     1
```